

Vegetation Change and Burned Area Patterns in Parque Nacional Peneda-Gerês

Using Landsat 8/9 and MODIS Burned Area Product

Study Area and Data

This study examines vegetation condition change in the Peneda-Gerês National Park, Portugal over a ten-year period from 2014 to 2024 using multi-temporal satellite imagery and remote sensing analysis. Vegetation change was compared with cumulative burned area to study long-term disturbance and recovery patterns within the park.

Parque Nacional Peneda-Gerês is Portugal's only national park and its drought-tolerant landscape is characterized by forests, shrublands, and mountainous terrain. The park experiences seasonal wildfire activity characteristic of Mediterranean ecosystems and comparable to the chaparral biome found in regions such as Southern California. The study was conducted using Google Earth Engine for analysis and QGIS for cartographic outputs.

Datasets: World Database on Protected Areas (WDPA), Landsat 8 & 9 (Collection 2 Level 2), MODIS Burned Area Product (MCD64A1 Version 6.1)

Methodology

Vegetation Condition

Vegetation Condition was determined using the Enhanced Vegetation Index (EVI), a satellite-derived metric used to assess vegetation greenness and photosynthetic activity. EVI was calculated from Landsat 8 and 9 Collection 2 Level 2 surface reflectance imagery:

$$EVI = 2.5 * ((NIR - RED) / (NIR + 6 * RED - 7.5 * BLUE + 1))$$

Cloud, cloud shadow, snow, and cirrus pixels were masked using the Landsat QA_PIXEL band. To reduce variability associated with atmospheric conditions and anomalous individual years, 2-year growing season mean EVI composites were created for an early study period (2014-2016) and a late study period (2022-2024). Imagery was restricted to the May-October growing season.

The EVI change map was created by subtracting the early-period mean EVI composite from the late-period mean EVI composite:

$$EVI\ Change = Mean\ EVI\ (2022-2024) - Mean\ EVI\ (2014-2016)$$

Positive values indicate higher vegetation condition during the later study period, while negative values indicate lower vegetation condition compared to the early study period.

Burned Area

Wildfire disturbance was analyzed using the MODIS MCD64A1 burned area product. Burned pixels were combined into a cumulative burned area layer representing all mapped burned areas during the study period.

Results

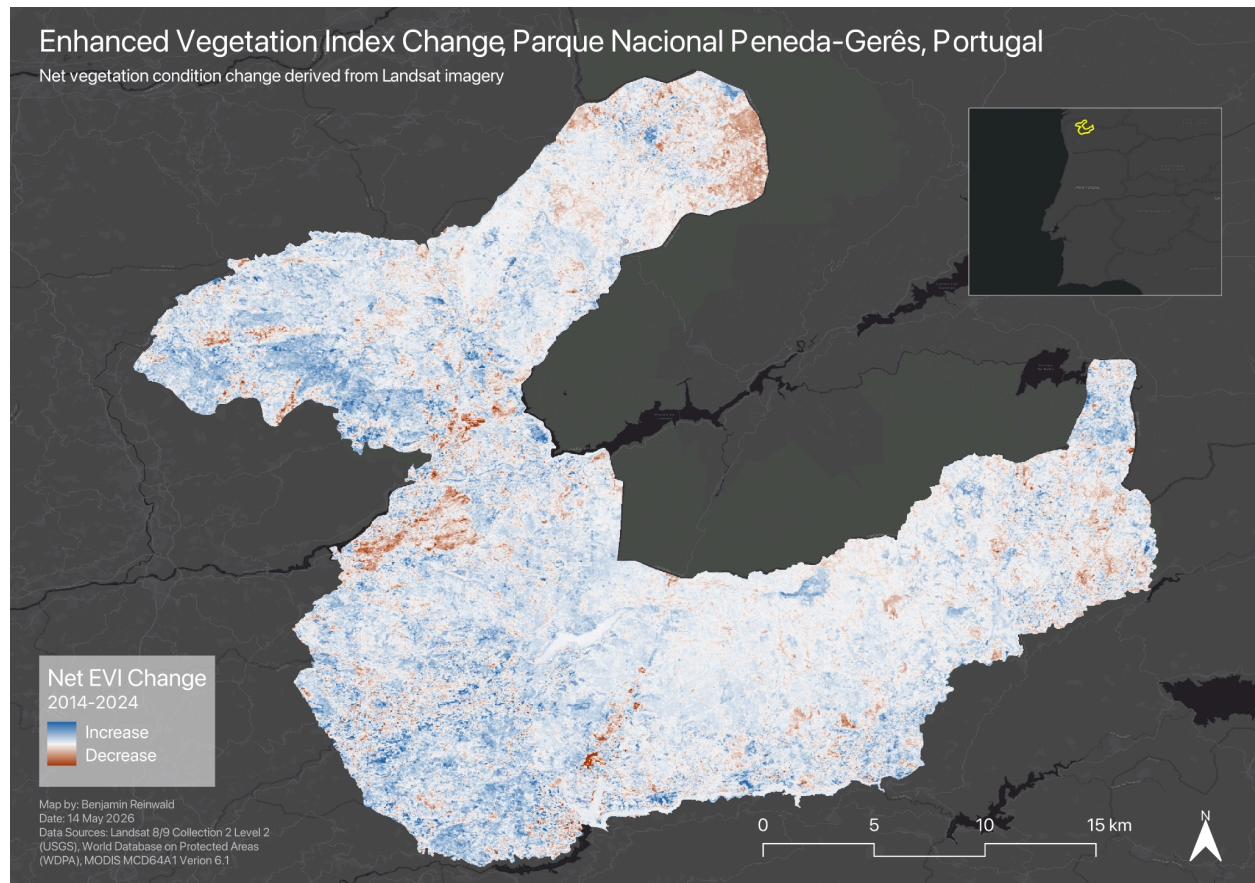


Figure 1. Net Enhanced Vegetation Index (EVI) change across Peneda-Gerês National Park between 2014 and 2024 derived from Landsat 8/9 imagery. Blue areas indicate long-term increases in vegetation condition and orange areas indicate long-term decreases in vegetation condition. Most of the park shows relatively stable or improving vegetation condition over the study period

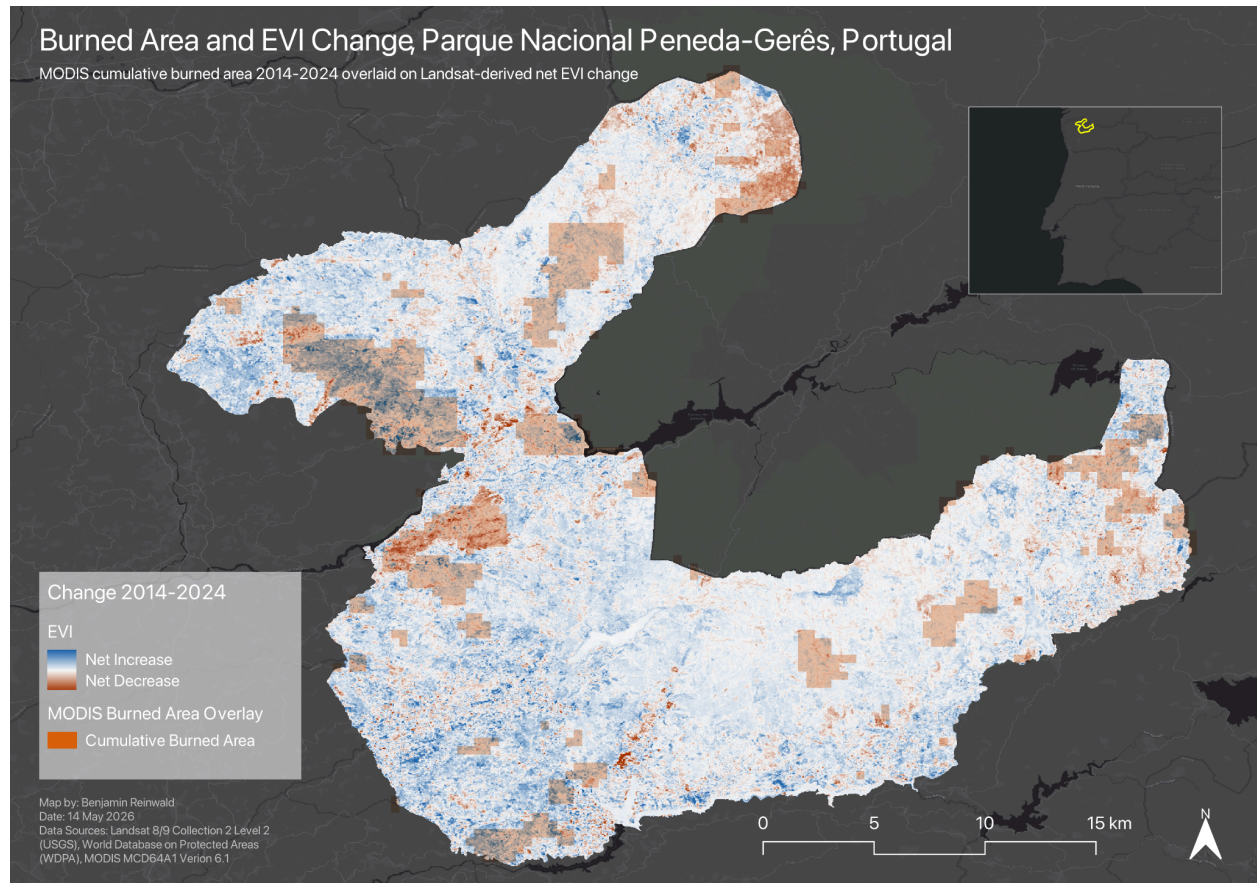


Figure 2. Cumulative MODIS burned area detections from 2014-2024 overlaid on Landsat-derived net EVI change during the study period. Burned areas frequently overlap with areas of both vegetation condition increases and decline; however, burned areas do not commonly overlap with areas of no significant change in vegetation condition.

Annual Burned Area and Mean EVI Trends, Peneda-Gerês (2014-2024)

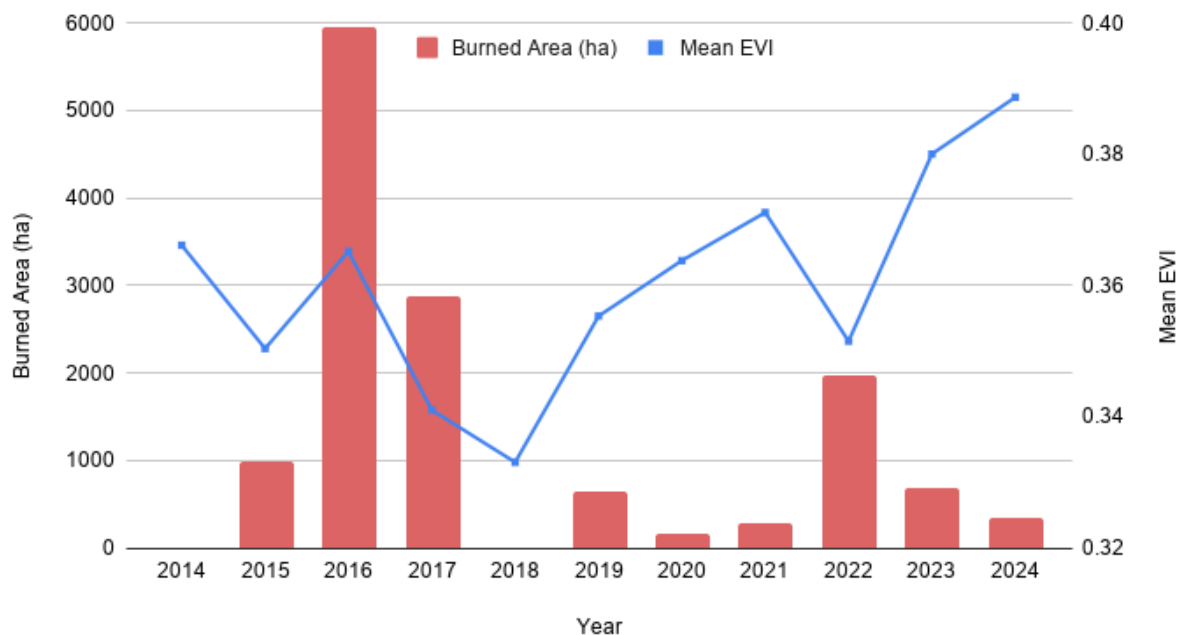


Figure 3. Annual burned area in hectares and mean EVI trends for Peneda-Gerês National Park from 2014-2024. Burned area peaked in 2016 and 2017, followed by a decrease in mean EVI until 2018. After these significant wildfire events, mean EVI increased until an additional decline and rebound during another major wildfire year in 2022.

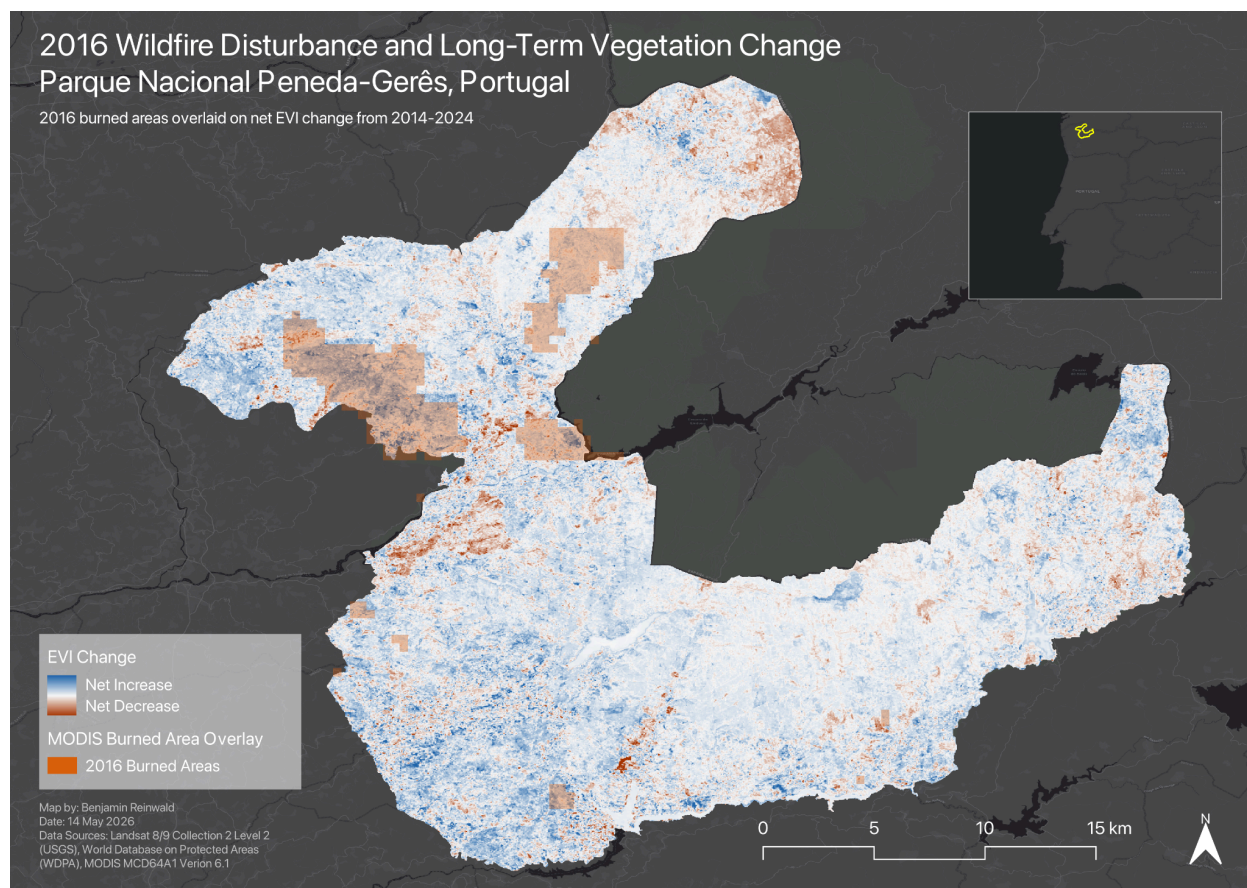


Figure 4. Burned areas from the 2016 wildfire season overlaid on cumulative EVI change from 2014-2024. Many areas affected by the 2016 wildfire show long-term increases in EVI. The early timing of the wildfires within the study period likely explains the observed vegetation recovery.

Discussion

Mean EVI increased by approximately 0.013 between the early (2014-2016) and late (2022-2024) study periods. Approximately 13,370 hectares of cumulative burned area were mapped using the MODIS burned area product.

Spatial comparison between the EVI change layer and the cumulative MODIS burned area layer suggests that wildfire disturbance was associated with localized vegetation decline. However, many burned areas also showed positive long-term EVI change during the later study period, pointing towards post-fire vegetation recovery or regrowth.

The annual growing season EVI and burned area chart revealed an interesting relationship during 2016, which shows the largest burned area of the study period. Despite this, annual mean EVI did not decline during the same year. Further analysis of the MODIS BurnDate layer showed that the 2016 fires occurred mostly between August and November. This likely explains why park-wide growing season mean EVI remained high during 2016 even with extensive burned area. Because annual EVI was calculated using May-October imagery, much of the vegetation captured during the 2016 growing season reflected pre-fire vegetation conditions. However, mean EVI declined during the two years following the fires, indicating

decreased vegetation condition after the disturbance event. EVI values began rebounding after 2018 and increased through 2022, when another major burned-area year occurred. Additionally, large portions of the areas affected by the 2016 wildfires show long-term increases in EVI by 2024, representing substantial post-fire vegetation recovery within this fire-adapted ecosystem. Because this significant burned-area event occurred relatively early in the study period, these areas had additional time for post-fire regrowth and recovery.

Overall, the results show that wildfire activity is associated with short-term vegetation decline followed by longer-term vegetation recovery in Peneda-Gerês National Park.